



JKBOSE 2026

CLASS 11TH

TRIGONOMETRIC FUNCTIONS

MATHS

LECTURE-03

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TOPICS TO BE COVERED

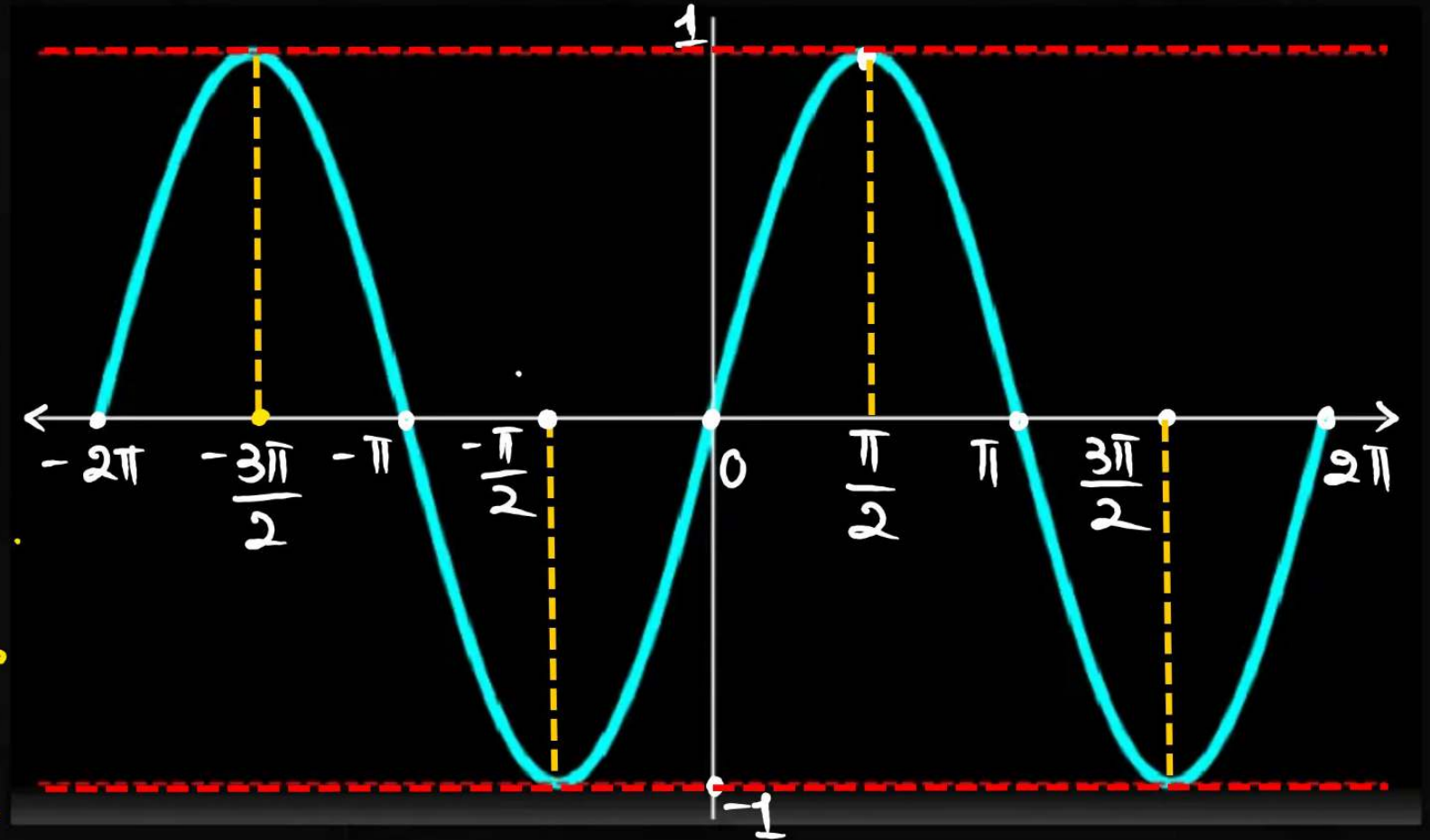
1 **Graphs of Trigonometric functions**

2 **Related Problems**

Graphs of Trigonometric Functions

Graph of $\sin x$:

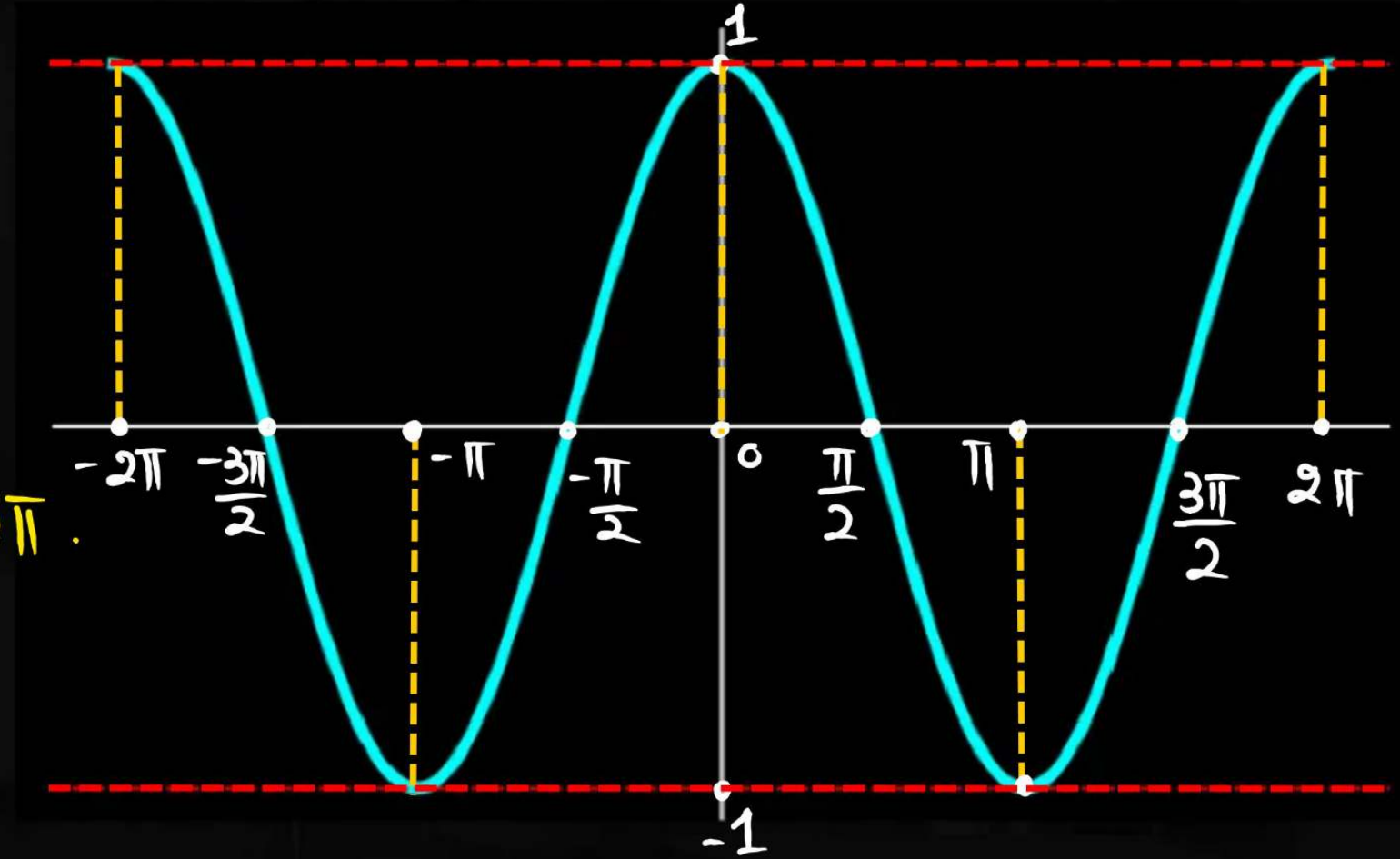
- $\sin x$ is defined for all $x \in \mathbb{R}$, so
Domain = $\mathbb{R}$.
- Range = $[-1, 1]$.
- $\sin x$ is a periodic function with period 2π .
- **1st Quadrant:** $\sin x$ increases from 0 to 1.
- **2nd Quadrant:** $\sin x$ decreases from 1 to 0.
- **3rd Quadrant:** $\sin x$ decreases from 0 to -1.
- **4th Quadrant:** $\sin x$ increases from -1 to 0.



Graphs of Trigonometric Functions

Graph of $\cos x$:

- Since $\cos x$ is defined for all $x \in \mathbb{R}$,
So Domain = $\mathbb{R}$.
- Range = $[-1, 1]$
- $\cos x$ is a periodic function with a period 2π .
- **1st Quadrant:** $\cos x$ decreases from 1 to 0.
- **2nd Quadrant:** $\cos x$ decreases from 0 to -1.
- **3rd Quadrant:** $\cos x$ increases from -1 to 0.
- **4th Quadrant:** $\cos x$ increases from 0 to 1.



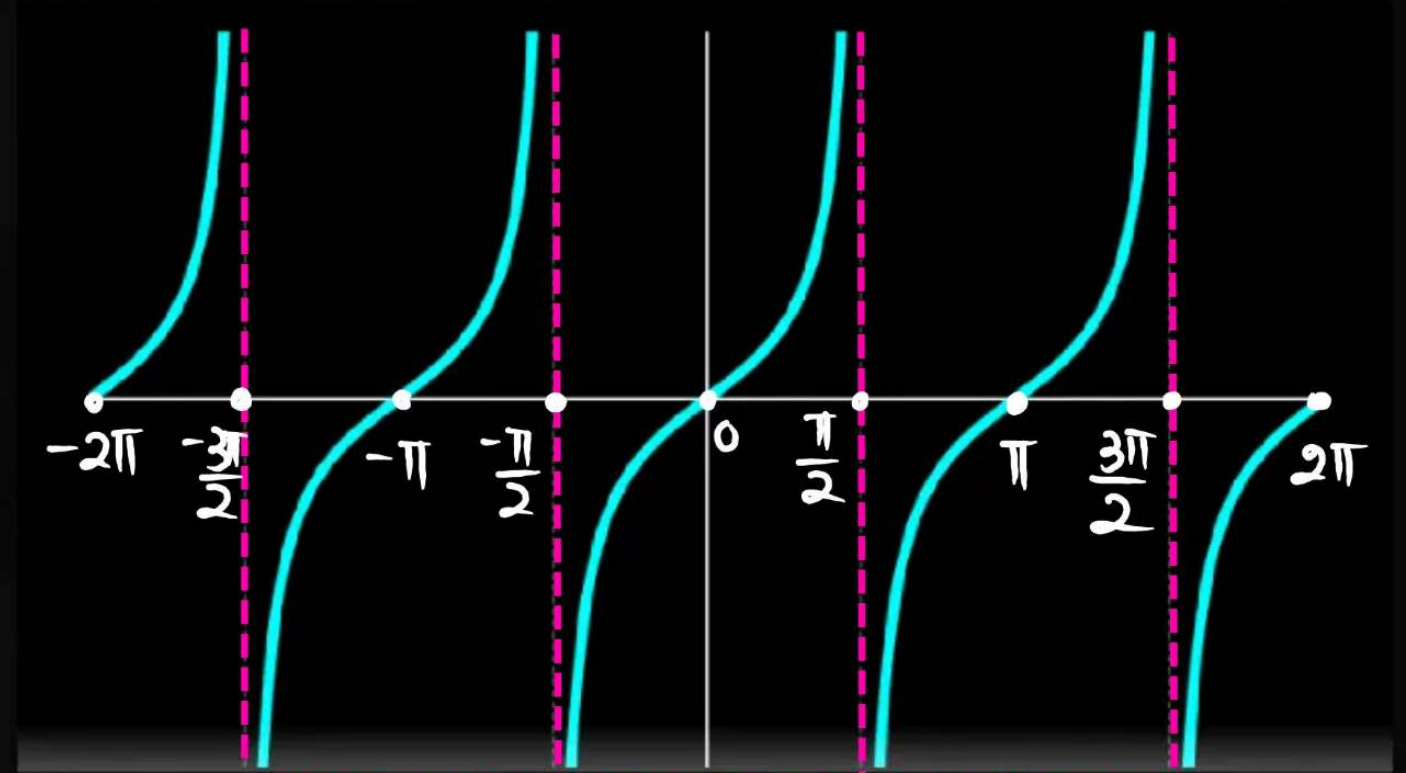
Graphs of Trigonometric Functions

Graph of $\tan x$:

We know that $\tan x = \frac{\sin x}{\cos x}$

Here $\cos x = 0 \Rightarrow x = (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$

- So, Domain = $\mathbb{R} - \left\{ (2n+1)\frac{\pi}{2} : n \in \mathbb{Z} \right\}$
- Range = $\mathbb{R}$
- **1st Quadrant:** $\tan x$ increases from 0 to ∞ .
- **2nd Quadrant:** $\tan x$ increases from $-\infty$ to 0.
- **3rd Quadrant:** $\tan x$ increases from 0 to ∞ .
- **4th Quadrant:** $\tan x$ increases from $-\infty$ to 0.



* $\tan x$ is a periodic function with a period π .

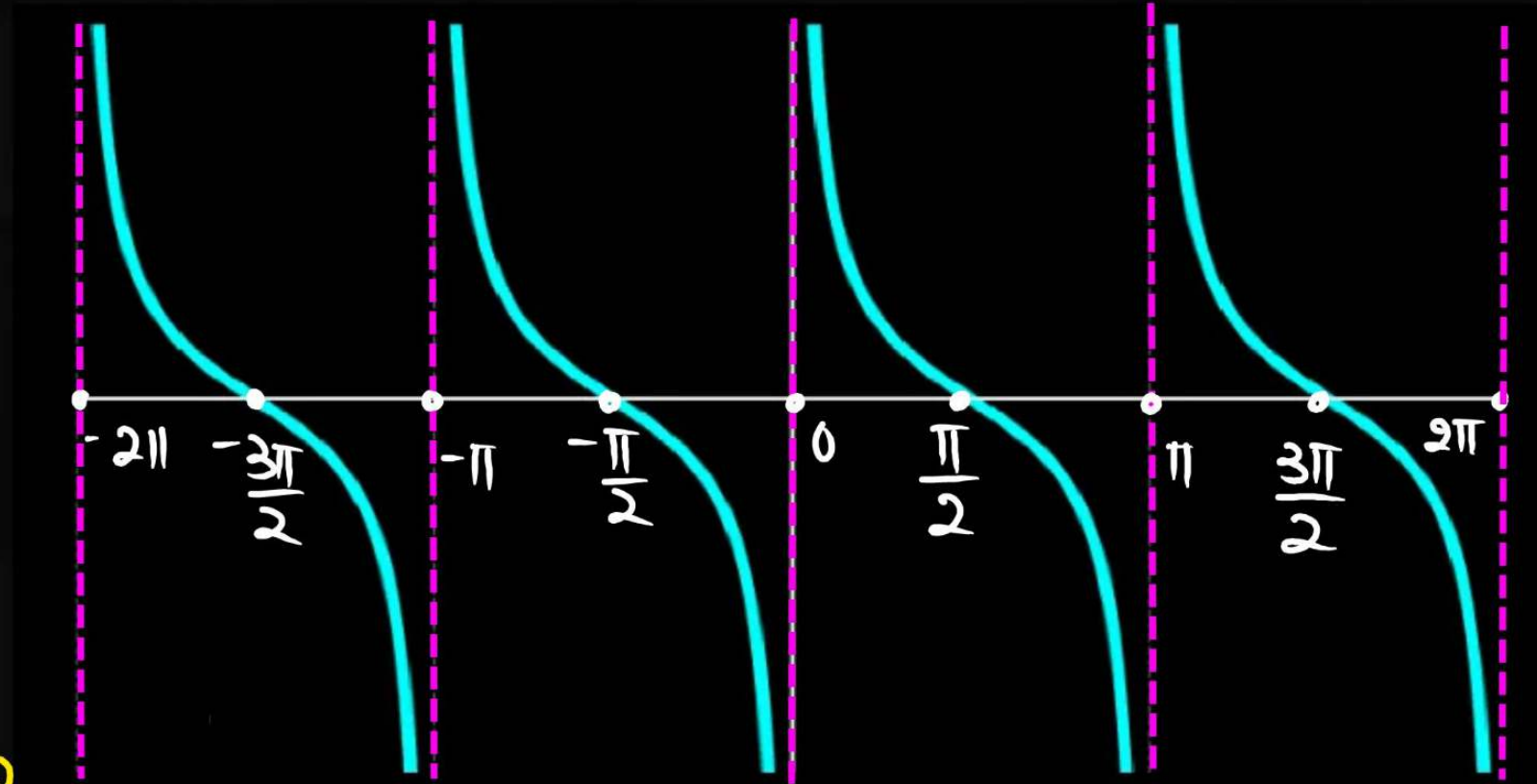
Graphs of Trigonometric Functions

Graph of $\cot x$:

We know that $\cot x = \frac{\cos x}{\sin x}$.

Here $\sin x = 0 \Rightarrow x = n\pi, n \in \mathbb{Z}$

- So, Domain = $\mathbb{R} - \{n\pi : n \in \mathbb{Z}\}$.
- Range = $\mathbb{R}$
- **1st Quadrant:** $\cot x$ decreases from ∞ to 0.
- **2nd Quadrant:** $\cot x$ decreases from 0 to $-\infty$.
- **3rd Quadrant:** $\cot x$ decreases from ∞ to 0.
- **4th Quadrant:** $\cot x$ decreases from 0 to $-\infty$.



$\cot x$ is a periodic function with a period π .

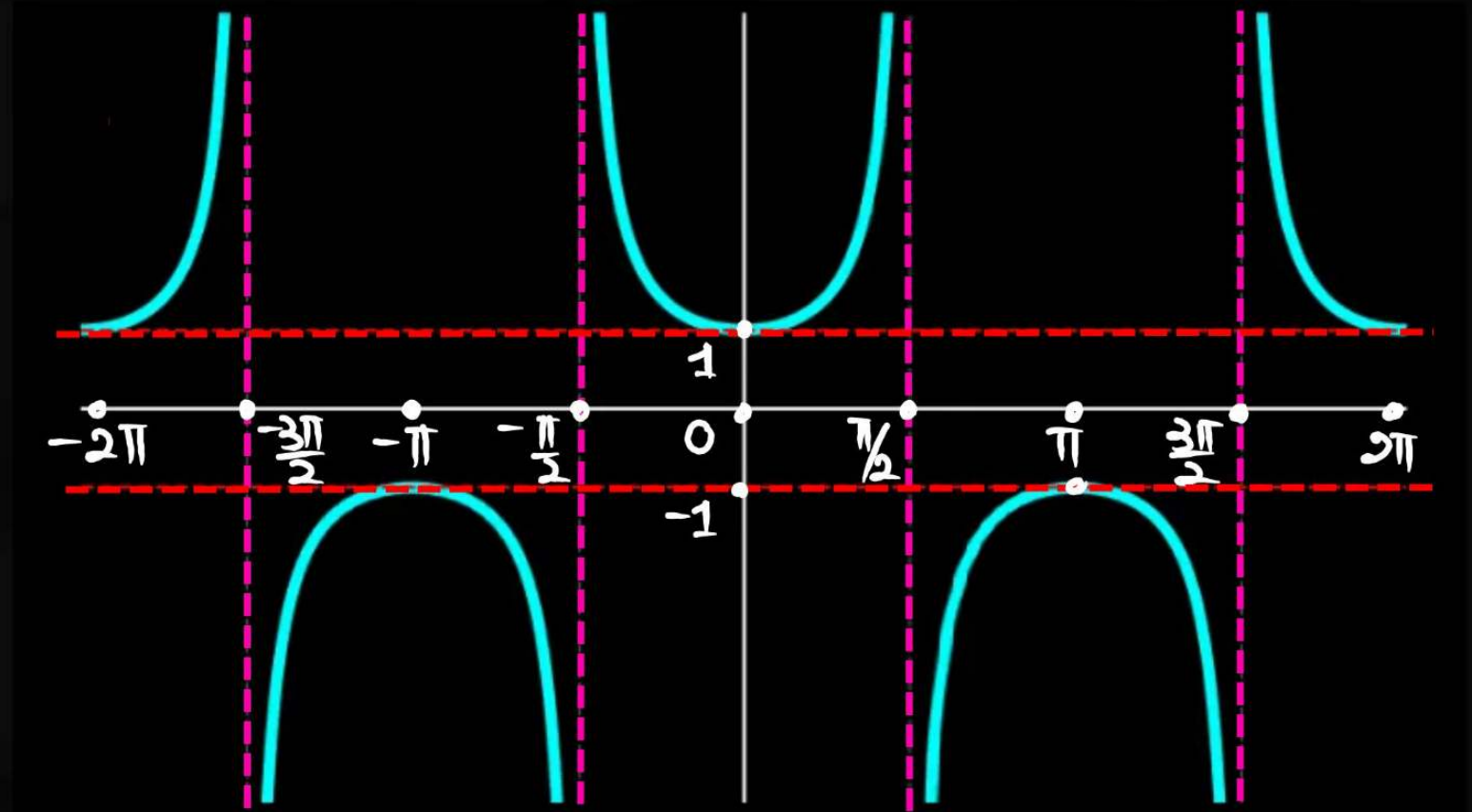
Graphs of Trigonometric Functions

Graph of $\sec x$:

We know that $\sec x = \frac{1}{\cos x}$

Here $\cos x = 0 \Rightarrow x = (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$

- So Domain = $\mathbb{R} - \left\{ (2n+1)\frac{\pi}{2} : n \in \mathbb{Z} \right\}$
- Range = $(-\infty, -1] \cup [1, \infty) = \mathbb{R} - (-1, 1)$
- 1st Quadrant: _____
- 2nd Quadrant: _____
- 3rd Quadrant: _____
- 4th Quadrant: _____



$\sec x$ is a periodic function with a period 2π .

Graphs of Trigonometric Functions

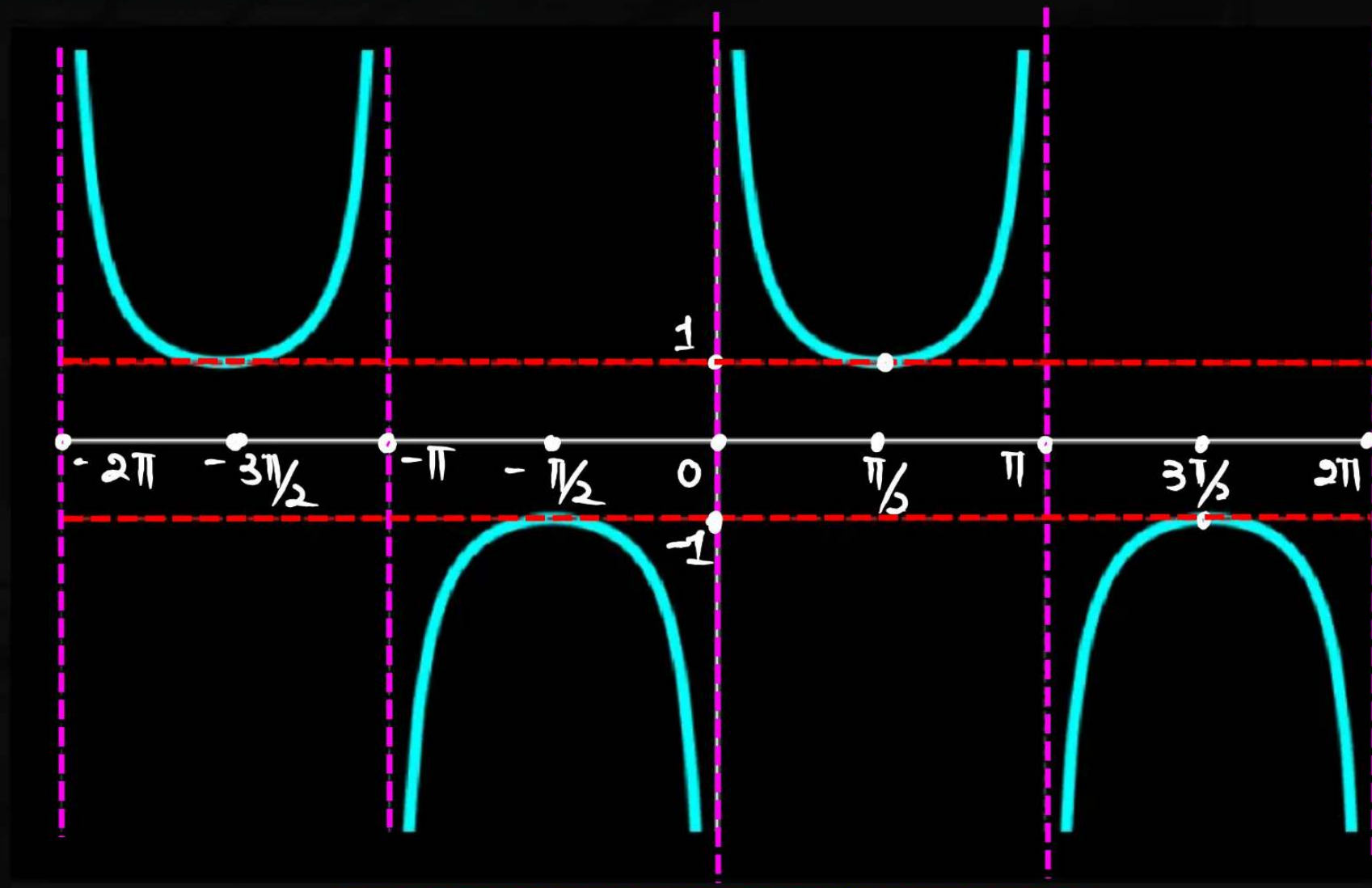
Graph of cosec x:

We know that $\text{cosec } x = \frac{1}{\sin x}$

Here $\sin x = 0 \Rightarrow x = n\pi, n \in \mathbb{Z}$

- So, Domain = $\mathbb{R} - \{n\pi : n \in \mathbb{Z}\}$
- Range = $(-\infty, -1] \cup [1, \infty) = \mathbb{R} - (-1, 1)$

- 1st Quadrant:
- 2nd Quadrant:
- 3rd Quadrant:
- 4th Quadrant:



cosec x is a periodic function with period 2π

QUESTION

$\tan x$ is defined for all x in

~~1~~

$\mathbb{R}$

~~2~~

$\mathbb{R} - \{n\pi: n \in \mathbb{Z}\}$

~~3~~

$\mathbb{R} - \left\{ (2n+1)\frac{\pi}{2}: n \in \mathbb{Z} \right\}$

~~4~~

None of these

QUESTION

cot x is defined for all x in

~~1~~

R

~~2~~

$R - \{n\pi: n \in \mathbb{Z}\}$

~~3~~

$R - \left\{(2n + 1)\frac{\pi}{2}: n \in \mathbb{Z}\right\}$

~~4~~

None of these

QUESTION

The function $f(x) = \tan x$ increases in

- ☒ 1 Ist and 2nd Quadrant
- ☒ 2 2nd and 3rd Quadrant
- ☒ 3 3rd Quadrant
- ☒ 4 All Quadrants

QUESTION

The function $f(x) = \cot x$ decreases in

- ☒ 1 Ist and 2nd Quadrant
- ☒ 2 2nd and 3rd Quadrant
- ☒ 3 3rd Quadrant
- ☒ 4 All Quadrants

QUESTION

The function $f(x) = \sin x$ decreases in

~~1~~

Ist and 2nd Quadrant

~~2~~

2nd and 3rd Quadrant

~~3~~

4th Quadrant

~~4~~

All Quadrants

QUESTION

The function $f(x) = \cos x$ decreases in

☒ 1 Ist and 2nd Quadrant

☐ 2 2nd and 3rd Quadrant

☐ 3 4th Quadrant

☐ 4 All Quadrants

QUESTION

The number of solutions of the equation $\tan x = |x|$ is

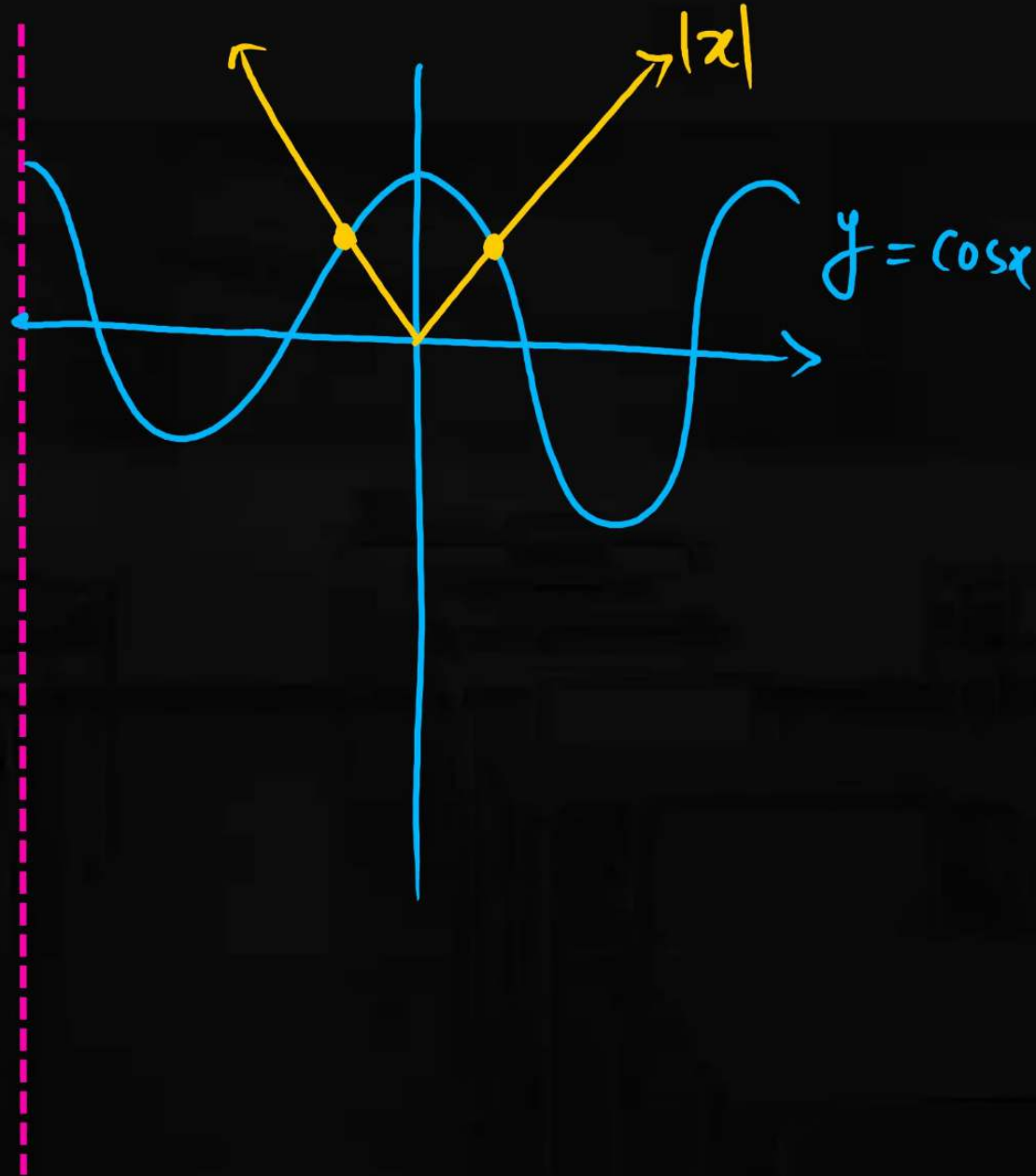
- ☒ One
- ☒ Two
- ☒ Three
- ☒ Infinite

Points of intersection of $\tan x$ and $|x|$.

QUESTION

The number of solutions of the equation $\cos x = |x|$ is

- ☒ One
- ☒ Two
- ☐ Three
- ☐ Zero



QUESTION

The value of $\tan 1^\circ \tan 2^\circ \tan 3^\circ \tan 4^\circ \dots \tan 89^\circ$ is equal to :

- 1** 0
- 2** 1
- 3** $\frac{1}{2}$
- 4** Not defined

(Home work)
Problem.

Thank You



Jan - mohdaj1729 ✓
✱